



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year: 2025-2026
ISE-1
Class: BTech (IT)

Date: 21/08/25
Time: 11:00 -11:45

Subject: Deep Learning
Code: PEC-IT407

Max Marks: 20

Notes:

1. All Questions are compulsory.
2. Figures to the right indicate full marks.

Q.N.	Question	Marks	CO	BT
1.	List and explain two real-world applications of deep learning.	[3]	PEC-IT407.1	BT2
2.	What role does the learning rate play in the backpropagation algorithm?	[3]	PEC-IT407.2	BT1
3	Compare and contrast the ReLU and Sigmoid activation functions.	[3]	PEC-IT407.2	BT4
4.	Write the mathematical expression for the Tanh activation function. State one advantage and one drawback.	[3]	PEC-IT407.2	BT3
5.	Derive the mathematical formulation of the backpropagation algorithm for a feedforward neural network. Clearly explain the role of chain rule in weight updates.	[3]	PEC-IT407.1	BT5
6	Consider a neural network with 2 inputs, one hidden layer with 2 neurons (using sigmoid activation), and one output neuron. Perform forward propagation for input $x = [1, 0]$, assuming initial weights = 0.5 and bias = 0.1 for all neurons. Show all calculations.	[5]	PEC-IT407.2	BT3

Course Outcomes (CO)

- PEC-IT407.1 Understand introduction application of deep learning.
PEC-IT407.2 To understand neurons, activation function.
PEC-IT407.3 To Understand optimizers.
PEC-IT407.4 To learn algorithms ANN, CNN
PEC-IT407.5 Understanding application and research level of the topics.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Information Technology
In-Semester Examination-I
Academic Year: 2024-25 (Semester – I)

Class: B.Tech (IT)	Subject: Mobile and Wireless Communication	Subject Code: PCC-IT402
Date: 18.08.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note: Attempt all questions.

Draw neat diagrams wherever necessary.

Solve any four Questions.

Q.1	What is ICAP? Explain it's Role and Data Flow in Content adaptation.	5	CO4	BT2,BT4
Q.2	Discuss the Architecture and Advantages of Interactive Voice Response System (IVRS).	5	CO2	BT4
Q.3	Compare TDMA and CDMA Techniques used in 2G and 3G.	5	CO2	BT1,BT2, BT3
Q.4	Explain the Bluetooth Protocol Stack with neat Diagram.	5	CO3	BT4
Q.5	Explain the Three tier architecture of mobile computing with examples	5	CO1	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Explain the fundamental physical and technical features of mobile and wireless Communications systems.
CO 2	Describe the fundamental principles of the mobile and wireless communications systems.
CO 3	Describe the implementation and development of mobile and wireless communication System.
CO 4	Interpret the information gathered and the outcomes of measurements.
CO 5	Conduct field tests and measurements of devices, equipment, actual components and system in the laboratory.
CO6	Test the technical features of mobile and wireless communication equipment.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Information Technology
In-Semester Examination I
Academic Year: 2025-26 (Semester – I)

Class: Final Year B. Tech.	Subject: Cryptography and Network Security	Subject Code: PCC-IT401
Date: 19.08.2025	Time: 11:00 to 11:45 am	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Use of non-programmable calculator is permitted.

Q.N.		Marks	CO	BT
Q.1	a) Encrypt and Decrypt the plaintext "ATTACK IS TONIGHT" with the help of Hill cipher using the key $K = \begin{pmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{pmatrix}$	10 M	CO2	BT5
Q.2	a) Answer the following questions using a Playfair cipher is given by its key K= LARKSPUR.			
	i) Encrypt the plaintext "SUNNY MOUNTAIN MEADOW"	05 M	CO1	BT3
	ii) Decrypt the cryptotext "ROCKY MOUNTAIN MEADOW"			
	b) Explain Active attack and Passive attack with suitable example?	05 M	CO1	BT2

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO1	Understand the concepts of classical encryption techniques and concepts of finite fields and number theory.
CO2	Explore working principles and utilities of various cryptographic algorithms including secret key cryptography, hashes and message digests, and public key algorithms.
CO3	Explore design issues and working principles of various authentication protocols, PKI standards.
CO4	Explore various secure communication standards including Kerberos, IPsec, and SSL/TLS and email.
CO5	Use existing cryptographic utilities to build programs for secure communication.